

# Artificial Intelligence in Healthcare Management and Economics

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*Evidence, Value, Governance, and Lifecycle Decision-Making*

## START HERE

*Digital Resource Companion*

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## Purpose of the companion

The Digital Resource Companion is the applied extension of Artificial Intelligence in Healthcare Management and Economics: Evidence, Value, Governance, and Lifecycle Decision-Making. The book contains the durable conceptual, economic, methodological, and governance argument. The companion provides editable institutional records, structured workbooks, worked examples, decision pathways, reference guides, process maps, and reproducible analytical resources that support application and local adaptation.

The companion does not replace the reasoning developed in the book. It provides a structured way to apply that reasoning to a clearly specified jurisdiction, organization, population, workflow, intervention, comparator, technology version, evidence base, and accountable decision process.

## Choose the appropriate entry point

| Entry point                          | Best suited to                                                                                  | What it provides                                                                                                                                            |
|--------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub Pages Web Companion           | Most users beginning online                                                                     | Guided navigation by chapter, decision stage, form, workbook, worked example, or analytical method.                                                         |
| Reader and Practitioner Edition      | Readers, practitioners, managers, analysts, educators, and institutional teams                  | A complete, accessible offline collection with the Web Companion, forms, workbooks, cases, guides, selected analytics, catalogues, and printable companion. |
| Core Tools Pack                      | Teams that primarily need the principal institutional documents                                 | The consolidated Core Resource Forms, consolidated Core Resource Workbook, practical guidance, licence, disclaimer, and checksums.                          |
| Canonical Source                     | Researchers, developers, repository maintainers, advanced analysts, educators, and institutions | The complete authoritative source collection, including editable, computational, machine-readable, process, publication, and website materials.             |
| Printable Digital Resource Companion | Readers who prefer a coherent readable reference                                                | A readable PDF explaining the architecture, resources, cases, methods, and recommended use; its editable Word source is retained in Canonical.              |

Most readers should begin with the Web Companion or Reader and Practitioner Edition. Both larger distributions contain the consolidated forms and workbook. The Core Tools Pack is supplied separately as a convenience subset, not as a nested archive or separate companion edition. Canonical retains the editable publication sources and full computational collection.

## Resource scope

| Resource class                 | V1.0 scope                                              |
|--------------------------------|---------------------------------------------------------|
| Book structure                 | 4 parts and 12 chapters                                 |
| Principal book figures         | 29                                                      |
| Book tables                    | 38, including the two front-matter tables               |
| Core institutional records     | 19                                                      |
| Worked examples                | 16, plus AIHE-CASE00 as the common case-analysis method |
| Python analytical modules      | 36                                                      |
| Catalogued companion resources | 126                                                     |

## The four core resource families

| Family                                   | Scope                                                                                                                         | Core identifiers     |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------|
| A — Framing and readiness                | Decision specification, participation, readiness, and data fitness.                                                           | AIHE-A01 to AIHE-A04 |
| B — Economics and modelling              | Economic specification, lifecycle cost, benefits, HTA, uncertainty, and reproducibility.                                      | AIHE-B01 to AIHE-B05 |
| C — Quality and institutional risk       | Quality control, corrective action, risk, human oversight, and communication.                                                 | AIHE-C01 to AIHE-C05 |
| D — Procurement and lifecycle governance | Supplier requirements, current decision states, monitoring, incidents, material change, renewal, replacement, and retirement. | AIHE-D01 to AIHE-D05 |

The AIHE prefix is retained as the stable resource identifier. Examples include AIHE-D02 for the decision-state record, AIHE-CASE12 for the model-update and renewal case, and AIHE-PY40 for advisory decision-state and trigger review.

## Recommended institutional sequence

1. Define the service problem, affected population, setting, intended use, accountable decision-maker, and realistic comparator.
2. Read the aligned book chapter and identify the smallest sufficient companion resource set.
3. Review the corresponding worked example and distinguish its synthetic assumptions from evidence needed locally.
4. Create a governed local copy of each selected record or workbook.
5. Replace illustrative content with authorized, versioned local evidence and preserve important unknowns or disagreements.
6. Complete multidisciplinary review proportionate to consequence, uncertainty, scale, reversibility, and institutional dependence.
7. Record the current decision state, conditions, mandatory safeguards, monitoring thresholds, transition triggers, accountable authority, next review date, and credible exit route.
8. Reconsider the decision after material change, new evidence, an incident, changed costs, renewal, restriction, replacement, or retirement.

## Using forms and workbooks together

| Word forms                                                                                                               | Excel workbooks                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Narrative justification, committee discussion, conditions, responsibilities, dissent, authorization, and sign-off.       | Structured data entry, costing, registers, comparisons, calculations, monitoring, and repeated review.                            |
| Use the consolidated volume when several institutional records are needed; choose an individual form for a focused task. | Use the consolidated workbook for several records; choose a specialist workbook or analytical template for a focused calculation. |
| Preserve the identifier, version, evidence cut-off, intervention identity, owner, and local changes.                     | Preserve the same controls, with data definitions, formula assumptions, input sources, and calculation checks.                    |

A form and workbook can be used together. The form preserves the narrative decision; the workbook holds structured supporting evidence. Neither replaces the current decision dossier.

## Decision lifecycle at a glance

The sequence is iterative. Later evidence or changed circumstances may reopen an earlier decision. Implementation activity remains distinct from institutional permission.

| Stage                                                     | Principal question                                                                                                                               | Typical core records                                |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1. Define the decision                                    | Is the problem sufficiently important and specified to justify further assessment?                                                               | AIHE-A01; AIHE-A02                                  |
| 2. Assess readiness and data                              | Are critical prerequisites adequate for the intended level of consequence and dependence?                                                        | AIHE-A03; AIHE-A04                                  |
| 3. Establish evidence and quality requirements            | Is the planned evidence proportionate to intended use, uncertainty, and consequence?                                                             | AIHE-C01; AIHE-C02; AIHE-EX05; AIHE-RG02; AIHE-RG06 |
| 4. Model cost, value, affordability, and HTA              | Does the intervention have a defensible value proposition and a financeable path under realistic scenarios?                                      | AIHE-B01; AIHE-B02; AIHE-B03; AIHE-B04              |
| 5. Select the decision model and characterize uncertainty | Is model complexity justified, credible, reproducible, and decision-relevant?                                                                    | AIHE-B05; AIHE-RG05                                 |
| 6. Assess governance and institutional risk               | Are mandatory safeguards satisfied and are residual risks explicitly owned and governable?                                                       | AIHE-C03; AIHE-C04; AIHE-C05; AIHE-RG04             |
| 7. Procure and contract                                   | Can the institution enforce the evidence, monitoring, change, continuity, and exit conditions required for safe use?                             | AIHE-D01                                            |
| 8. Implement, monitor, and control change                 | Does real-world use remain within approved performance, safety, workload, equity, cost, and governance conditions?                               | AIHE-D02; AIHE-D03; AIHE-D04; AIHE-D05; AIHE-RG03   |
| 9. Reassess continuing value, affordability, and support  | Does continuing use remain justified, affordable, supportable, and preferable to current alternatives without compromising mandatory safeguards? | AIHE-B02; AIHE-B03; AIHE-B04; AIHE-D01; AIHE-D05    |
| 10. Assemble the decision record and renew or retire      | Is continuing use still preferable to redesign, restriction, replacement, or retirement under current evidence and alternatives?                 | AIHE-D02; AIHE-D05; AIHE-P; AIHE-AE                 |

## Worked examples and computational resources

The sixteen worked examples and AIHE-CASE00 common method connect service problems with evidence, economics, governance, implementation, monitoring, and lifecycle decisions. Their scenarios and demonstration datasets are synthetic unless explicitly stated; they are not empirical evidence for a local decision.

The Canonical Source contains thirty-six Python analytical modules. Supplementary worked cases address financial recognition and budget incidence, the expected value of additional evidence, and upstream model-provider change. Method choice, versioned inputs, verification, validation, and expert interpretation remain necessary.

## Version control and responsible use

- Record the stable identifier, exact resource version, release date, access date, evidence cut-off, intervention version, and local modifications.
- Preserve the exact data, code, environment, assumptions, and outputs used in consequential analysis. Verify the currency of law, standards, software, and supplier information before reliance.
- A completed form, score, or model result does not authorize use. Mandatory safeguards remain non-compensatory, and consequential transitions require accountable human authority.
- Do not enter patient identifiers, credentials, private keys, confidential contracts, or restricted institutional information into public demonstrations or uncontrolled files.
- Maintain a credible route for restriction, rollback, replacement, retirement, and responsible closure.

The companion is an author-developed educational and institutional decision-support collection. It does not replace clinical, legal, regulatory, ethical, privacy, cybersecurity, procurement, accreditation, accounting, financial, or health technology assessment review.

## Access and citation

| Resource                                                         | Address                                                                                                                                                                         |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Persistent companion record (Concept DOI identified in the book) | <a href="https://doi.org/10.5281/zenodo.21753927">https://doi.org/10.5281/zenodo.21753927</a>                                                                                   |
| Canonical repository recorded in the source package              | <a href="https://github.com/ai-healthcare-management-economics/digital-resource-companion">https://github.com/ai-healthcare-management-economics/digital-resource-companion</a> |
| Web Companion recorded in the source package                     | <a href="https://ai-healthcare-management-economics.github.io/digital-resource-companion/">https://ai-healthcare-management-economics.github.io/digital-resource-companion/</a> |

This package is V1.0. The persistent record is not a newly assigned version-specific DOI. When the package is published, cite the version-specific DOI shown on that archived release and retain the exact resource identifier and local changes.